



Linear Models I: Introduction

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Stat 141 | Fall 2026
Wednesday, Week 4

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Week 4 Survey

- Fill out the **Week 4 Survey**



<https://tinyurl.com/stat141-4wk-survey>

05:00

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Goals for Today

- Discuss the ideas of statistical modeling
- Learn two new summary statistics
- Introduce simple linear regression

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Typical Analysis Goals

Descriptive: Want to estimate quantities (summary statistics) related to the population.

→ *How many trees are in the Amazon?*

Predictive: Want to predict the value of a variable.

→ *Can I use remotely sensed data to predict forest types in the Amazon?*

Causal: Want to determine if changes in a variable cause changes in another variable.

→ *Do financial contracts prevent people from deforesting their land in the Amazon?*

We will focus mainly on **descriptive modeling** in this course, and occasionally on **predictive modeling** (take Stat 243: Statistical Learning to learn more). If you want to learn more about **causality**, take Stat 394: Causal Inference.

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Form of the Model

$$y = f(x) + \epsilon$$

where ϵ represents an error term.

Goal:

- Determine a reasonable form for $f()$. (Ex: Line, curve, ...)
- Estimate $f()$ with $\hat{f}()$ (“f hat”) using the data.
- Generate predicted values: $\hat{y} = \hat{f}(x)$.

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Simple Linear Regression Model

Consider this model when:

- Response variable (y): quantitative
- Explanatory variable (x): quantitative
 - Have only ONE explanatory variable.
- AND, $f()$ can be approximated by a line.

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Example: Major League Baseball Data from the 1986 and 1987 seasons

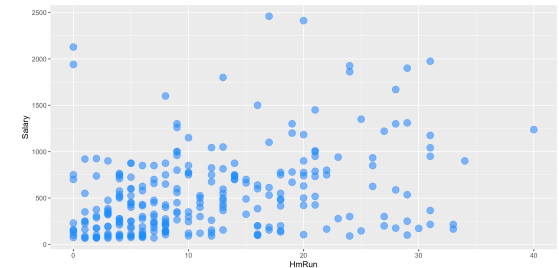
```
1 library(ISLR)
2 data(Hitters)
3 hitters <- Hitters %>% filter(!is.na(Salary), !is.na(HmRun))
4
5 glimpse(hitters)

Rows: 263
Columns: 20
$ AtBat <int> 315, 479, 496, 321, 594, 185, 298, 323, 401, 574, 202, 418, ...
$ Hits <int> 81, 130, 141, 87, 169, 37, 73, 81, 92, 159, 53, 113, 60, 43, ...
$ HmRun <int> 7, 18, 20, 10, 4, 1, 0, 6, 17, 21, 4, 13, 0, 7, 20, 2, 0, 16, ...
$ Runs <int> 24, 66, 65, 39, 74, 23, 24, 26, 49, 107, 31, 48, 30, 29, 89, ...
$ RBI <int> 38, 72, 78, 42, 51, 8, 24, 32, 66, 75, 26, 61, 11, 27, 75, 8, ...
$ Walks <int> 39, 76, 37, 30, 35, 21, 7, 0, 65, 59, 27, 47, 22, 30, 73, 15, ...
$ Years <int> 14, 3, 11, 2, 11, 2, 3, 2, 13, 10, 9, 4, 6, 13, 15, 5, 0, 1, ...
$ CatBat <int> 3449, 1624, 5628, 396, 4408, 214, 509, 341, 5206, 4631, 1876, ...
$ CHits <int> 835, 457, 1575, 101, 1133, 42, 108, 86, 1332, 1300, 467, 392, ...
$ ChmRun <int> 69, 63, 225, 12, 19, 1, 0, 6, 253, 90, 15, 41, 4, 36, 177, 5, ...
$ CRuns <int> 321, 224, 828, 48, 501, 30, 41, 32, 784, 702, 192, 205, 309, ...
$ CRBI <int> 414, 266, 838, 46, 336, 9, 37, 34, 890, 504, 186, 204, 103, ...
```

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Example: Major League Baseball Data from the 1986 and 1987 seasons

- Linear trend?
- Direction of trend?



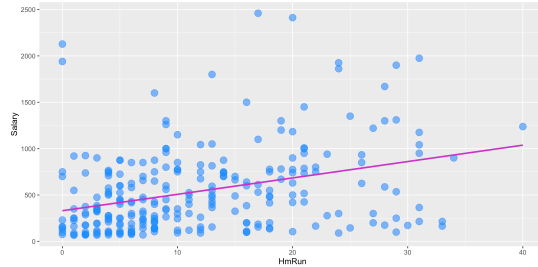
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Example: Major League Baseball Data from the 1986 and 1987 seasons

- A simple linear regression model could represent this data:

$$y = \beta_0 + \beta_1 x + \epsilon$$

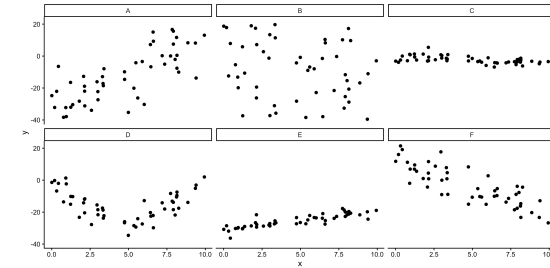
- β_0 and β_1 are fixed numbers (“coefficients”)
- β_0 represents the **intercept** of a line
- β_1 represents the **slope** of a line
- Knowing β_0 and β_1 would help us summarize and describe the relationship between x and y .



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Finding the Right Coefficients

- We want to find β_1 (slope) and β_0 (intercept) so that the line fits our data well
 - Need summary statistics that quantify the strength and relationship of the linear trend
 - These will help us find a value for slope and determine how well a line fits our data



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(Sample) Correlation Coefficient

- Measures the **strength** and **direction** of **linear** relationship between two quantitative variables
- Symbol: r
- Always between -1 and 1
- Sign indicates the direction of the relationship
- Magnitude indicates the strength of the linear relationship

r is calculated using the sample means ($\bar{x}$, $\bar{y}$) and standard deviations (s_x , s_y) of the variables x and y :

$$r = \frac{1}{s_x s_y} \cdot \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

```
1 cor(hitters$HitsRun, hitters$Salary)
[1] 0.3430281
```

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Sidenote: (Sample) Covariance

Sample Correlation Coefficient:

$$r = \frac{1}{s_x s_y} \cdot \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

Sample Covariance:

$$\text{cov}(x, y) = r \times s_x s_y = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

The sample correlation coefficient is a **standardized sample covariance**, which is what causes it to only take values from -1 to 1. The sample covariance can take any real value.

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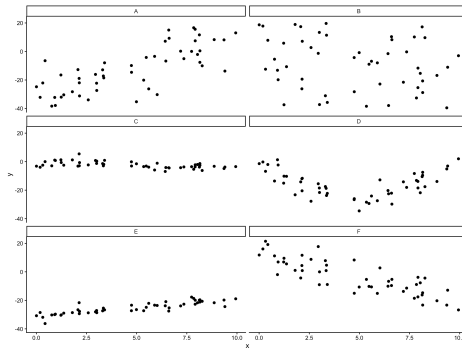
Correlation, Covariance, and Slope

Q: Which will have the largest *positive* correlation?

Q: Which will have the largest *negative* correlation?

Q: Which will have correlation closest to 0?

A: 0.7568 B: -0.2172 C: -0.5373 D: -0.1133
E: 0.863 F: -0.8343



- Correlation: how strong is the linear relationship *relative to noise*
- Correlation and slope are related, but not the same (ex. A vs E)

02:00

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New Example

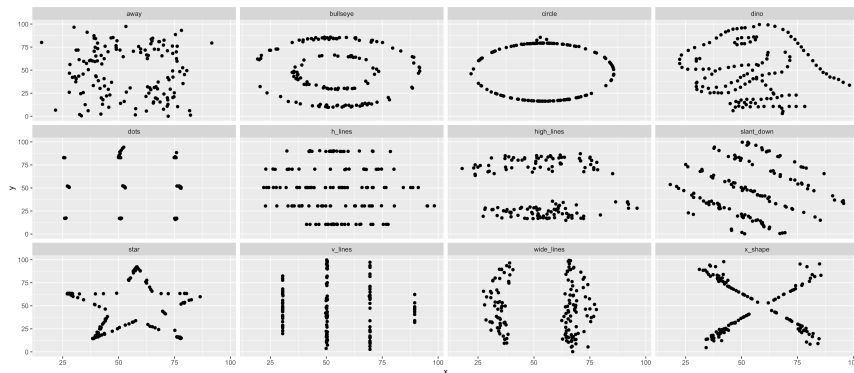
```
1 # Correlation coefficients
2 dat2 %>%
3   group_by(dataset) %>%
4   summarize(cor = cor(x, y))

# A tibble: 12 x 2
  dataset      cor
  <chr>      <dbl>
1 away      -0.0641
2 bullseye   -0.0686
3 circle     -0.0683
4 dino       -0.0645
5 dots       -0.0603
6 h_lines    -0.0617
7 high_lines -0.0685
8 slant_down -0.0690
9 star       -0.0630
10 v_lines    -0.0694
11 wide_lines -0.0666
12
```

- Conclude that x and y have the same relationship across these different datasets because the correlation is the same?

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Always graph the data when exploring relationships!



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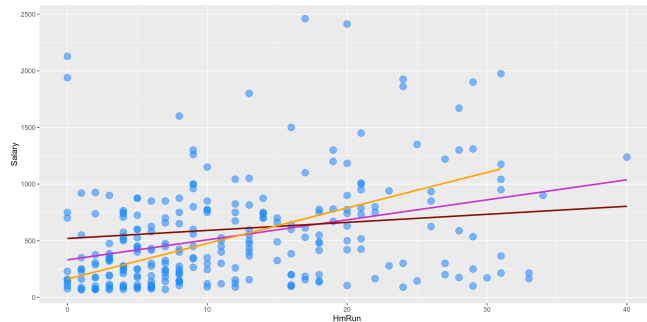
Returning to the Simple Linear Regression model...

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Simple Linear Regression

Let's return to the MLB hitters example.

- A line is a reasonable model form.
- Where should the line be? Slope? Intercept?



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Form of the SLR Model

$$y = f(x) + \epsilon$$

$$y = \beta_0 + \beta_1 x + \epsilon$$

- We assume y is not perfectly predicted by $f(x)$, with ϵ representing random error.
- Need to determine the best **estimates** of β_0 and β_1 .

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Distinguishing between the population and the sample: Terminology

$$y = \beta_0 + \beta_1 x + \epsilon$$

Population model

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

Fitted or estimated model

- | | |
|--|--|
| <ul style="list-style-type: none"> • Parameters: <ul style="list-style-type: none"> ■ Based on the population ■ Unknown if we don't have data on the whole population ■ EX: β_0 and β_1, AKA population coefficients | <ul style="list-style-type: none"> • Statistics: <ul style="list-style-type: none"> ■ Based on the sample data ■ Known ■ Usually estimate a population parameter ■ EX: $\hat{\beta}_0$ and $\hat{\beta}_1$, AKA estimated or "fitted" coefficients ■ No ϵ over here! |
|--|--|

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Method of Least Squares: More Terminology

Need two key definitions:

- **Fitted value**: The *estimated* value of the i -th case, given estimates $\hat{\beta}_0$ and $\hat{\beta}_1$

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

- **Residuals**: The *observed* error term for the i -th case, given fitted values

$$e_i = y_i - \hat{y}_i$$

Goal: Pick values for $\hat{\beta}_0$ and $\hat{\beta}_1$ so that the residuals are small!!

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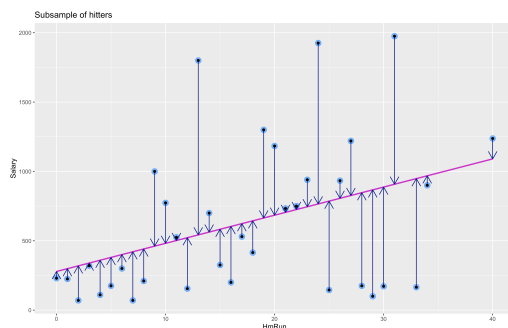
Method of Least Squares: Minimizing Residuals

- Want residuals to be small.
- Minimize a function of the residuals.
- Minimize:

$$\sum_{i=1}^n e_i^2$$

Sidenote:

- We could use $\sum_{i=1}^n |e_i|$
- But, this is less common, less computationally efficient, and lacks theoretical advantages
- $\sum_{i=1}^n e_i^2$ appropriately weights one large residual as “worse” than many small ones



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Method of Least Squares: Coefficient Formulas

- Suppose n observations of x and y are collected: $(x_1, y_1), \dots, (x_n, y_n)$.
- It turns out there are specific values of $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize the sum of squared residuals, given this data:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i \quad \text{and} \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

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Method of Least Squares: Coefficient Formulas

- Suppose n observations of x and y are collected: $(x_1, y_1), \dots, (x_n, y_n)$.
- It turns out there are specific values of $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize the sum of squared residuals, given this data:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{s_y}{s_x} r$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i \quad \text{and} \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

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Method of Least Squares: Interpretation

Once we know $\hat{\beta}_0$ and $\hat{\beta}_1$, we can estimate the whole function with:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

Called the **least squares line**, **regression line**, or the **line of best fit**.

We need to be precise and careful when interpreting β_0 and β_1 (and their estimates)

- **Intercept:** We [expect/predict] y to be β_0 on average when $x = 0$.
- **Slope:** For a one-unit increase in x , we [expect/predict] y to change by β_1 units on average.

I will be a stickler about this! Make sure to adhere to this language, using problem context, for full points.

Unless experimental data is involved, avoid causal language.

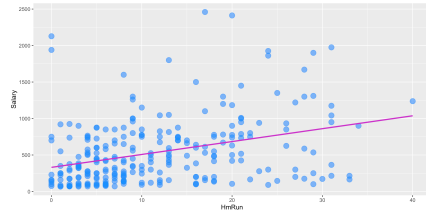
- Example of causal language: “When x increases by 1 unit, y will increase by β_1 ”

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Method of Least Squares: Plotting Regression Lines

`ggplot2` will compute the line and add it to your plot using `geom_smooth(method = "lm")`

```
1 ggplot(data = hitters,  
2       mapping = aes(x = HmRun,  
3                     y = Salary)) +  
4   geom_point(alpha = 0.6, size = 4,  
5             color = "dodgerblue") +  
6   geom_smooth(method = "lm", se = FALSE,  
7             color = "magenta3")
```



We can calculate the **exact** values of $\hat{\beta}_0$ and $\hat{\beta}_1$ using our formulas, by hand or in R

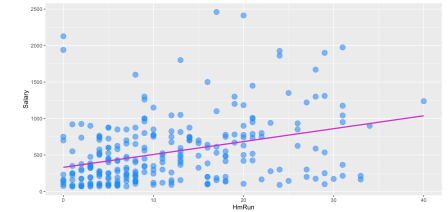
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Method of Least Squares: Practice Interpreting

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

Here, $\hat{\beta}_0 = 330.59$ and $\hat{\beta}_1 = 17.67$ (units of \$1000).

Q: How should we interpret these coefficients?



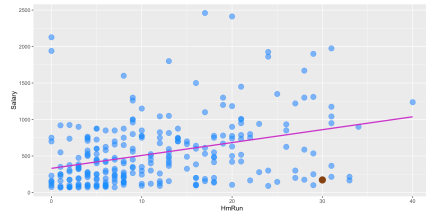
- **Intercept:** We expect/predict a player's **Salary** to be \$330,000 on average when they've hit 0 home runs.
- **Slope:** For **each additional home run** that a player hits, we expect/predict their **Salary** to increase by \$17,670 on average.

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Method of Least Squares: Practice Predicting

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

Here, $\hat{\beta}_0 = 330.59$ and $\hat{\beta}_1 = 17.67$ (units of \$1000).



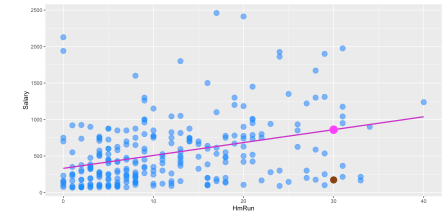
- **Q:** Suppose a player hit **HmRun = 30**. What does the model predict for **Salary**?

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Method of Least Squares: Practice Predicting

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

Here, $\hat{\beta}_0 = 330.59$ and $\hat{\beta}_1 = 17.67$ (units of \$1000).



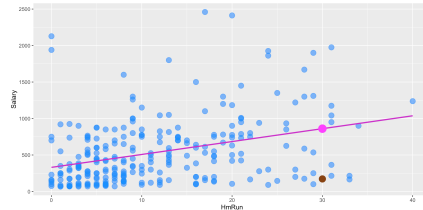
- **Q:** Suppose a player hit **HmRun = 30**. What does the model predict for **Salary**?
- **A:** $\hat{y} = 330.59 + 17.67 \cdot 30 = 860.69$
- This is different from the actual **Salary** we see for a player with **HmRun = 30**.
- This isn't unexpected: the line predicts the **average y** for each value of **x**.

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Method of Least Squares: Equations in LaTeX

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

Here, $\hat{\beta}_0 = 330.59$ and $\hat{\beta}_1 = 17.67$ (units of \$1000).



- **Q:** Suppose a player hit **HmRun = 30**. What does the model predict for **Salary**?
- **A:** $\hat{y} = 330.59 + 17.67 \cdot 30 = 860.69$

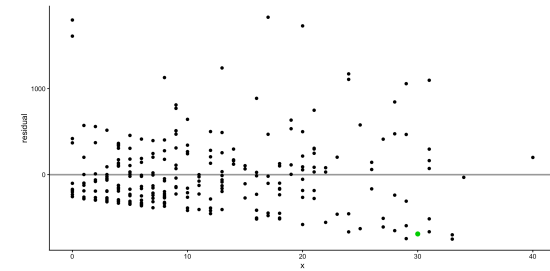
Side note: You will often be asked to write out this kind of model equation. You can do this nicely using a little LaTeX in your `.qmd` file, like this:

`$\widehat{y} = 330.59 + 17.67 \cdot 30$`

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Residual plot preview

We can visualize the accuracy of a linear model using residual plots:

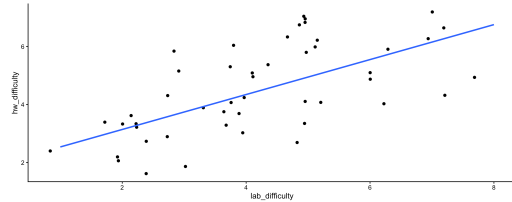


- We'll learn to use different residual plots to more formally assess linear model assumptions

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Activity: Extra practice interpreting coefficients

```
1 x <- read.csv("data/qods-anon.csv")
2 names(x) <- c("lab_difficulty", "hw_difficulty")
3
4 set.seed(123)
5 ggplot(x, aes(x = lab_difficulty, y = hw_difficulty)) +
6   geom_jitter() +
7   geom_smooth(method = "lm", se = FALSE) +
8   theme_classic()
```



```
1 coef(lm(hw_difficulty ~ lab_difficulty, x))
   (Intercept) lab_difficulty
         1.9365233          0.6022233
```

- **Q1:** How should we interpret the intercept?
- **Q2:** How should we interpret the coefficient on **lab_difficulty**?

03:00

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Recap

We considered the simple linear model:

$$y = \beta_0 + \beta_1 x + \epsilon$$

- β_0 and β_1 are fixed numbers (“coefficients”)
- β_0 represents the **intercept** of a line
- β_1 represents the **slope** of a line

We estimate these population coefficients using sample data via the **method of least squares**, which finds $\hat{\beta}_0$ and $\hat{\beta}_1$ such that the **sum of squared residuals is minimized**.

- Knowing $\hat{\beta}_0$ and $\hat{\beta}_1$ allows us to **predict** values of y given specific values of x
- There is a precise way that we should interpret $\hat{\beta}_0$ and $\hat{\beta}_1$ (see slides)

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Next time: more simple linear regression

- We'll use **R** to compute the least squares line and the values of $\hat{\beta}_0$ and $\hat{\beta}_1$ (we'll start practicing in lab).
- We'll introduce the formal assumptions we must make when doing linear regression, and diagnostic plots to check these assumptions.